

Question 1. [5 Marks]

a) Using Hidden Markov model to disambiguate the POS tag for the word “play” in the following sentence, given the transition probabilities and emission probabilities below: [5 marks]

- “He likes to play”
- Emission probabilities

	He	likes	to	play
PRP	0.8	0.2	0	0
VB	0	0.8	0	0.2
TO	0	0	1	0
NN	0.2	0	0	0.8

- Transition probabilities

	VB	TO	NN	PRP
<s>	0.2	0	0.2	0.6
VB	0.0	0.3	0.4	0.3
TO	0.4	0	0.5	0.1
NN	0.4	0.1	0.1	0
PRP	0.9	0.0	0.1	0

Solution:

Start by calculating the probabilities for each POS tag (PRP, VB, TO, NN) based on the start state (<s>)

$$V(\text{PRP}, "He") = P(\text{PRP}) \times P("He"|\text{PRP}) = 0.6 \times 0.8 = 0.48$$

$$V(\text{VB}, "He") = P(\text{VB}) \times P("He"|\text{VB}) = 0.2 \times 0 = 0$$

$$V(\text{TO}, "He") = P(\text{TO}) \times P("He"|\text{TO}) = 0 \times 0 = 0$$

$$V(\text{NN}, "He") = P(\text{NN}) \times P("He"|\text{NN}) = 0.2 \times 0.2 = 0.04$$

Now, calculate the probabilities for "likes" using the results from "He"

For PRP,

$$\text{From PRP: } 0.48 \times 0 \times 0.2 = 0$$

$$\text{From VB: } 0 \times 0.3 \times 0.2 = 0$$

$$\text{From TO: } 0 \times 0.1 \times 0.2 = 0$$

$$\text{From NN: } 0.04 \times 0 \times 0.2 = 0$$

Max for PRP = 0

For VB,

$$\text{From PRP: } 0.48 \times 0.9 \times 0.8 = 0.3456$$

$$\text{From VB: } 0 \times 0 \times 0.8 = 0$$

$$\text{From TO: } 0 \times 0.4 \times 0.8 = 0$$

$$\text{From NN: } 0.04 \times 0.4 \times 0.8 = 0.0128$$

Max for VB = 0.3456 (From PRP)

For TO,

$$\text{From PRP: } 0.48 \times 0 \times 0 = 0$$

$$\text{From VB: } 0 \times 0.3 \times 0 = 0$$

$$\text{From TO: } 0 \times 0 \times 0 = 0$$

$$\text{From NN: } 0.04 \times 0.1 \times 0 = 0$$

Max for TO = 0

For NN,

$$\text{From PRP: } 0.48 \times 0.1 \times 0 = 0$$

$$\text{From VB: } 0 \times 0.4 \times 0 = 0$$

$$\text{From TO: } 0 \times 0.5 \times 0 = 0$$

$$\text{From NN: } 0.04 \times 0.1 \times 0 = 0$$

Max for NN = 0

Then, we calculate the probabilities for "to"

For PRP,

From PRP: $0 \times 0 \times 0 = 0$

From VB: $0.3456 \times 0.3 \times 0 = 0$

From TO: $0 \times 0.1 \times 0 = 0$

From NN: $0 \times 0 \times 0 = 0$

Max for PRP = 0

For VB,

From PRP: $0 \times 0.9 \times 0 = 0$

From VB: $0.3456 \times 0 \times 0 = 0$

From TO: $0 \times 0.4 \times 0 = 0$

From NN: $0 \times 0.4 \times 0 = 0$

Max for VB = 0

For TO,

From PRP: $0 \times 0 \times 1 = 0$

From VB: $0.3456 \times 0.3 \times 0.1 = 0.10368$

From TO: $0 \times 0 \times 1 = 0$

From NN: $0 \times 0.5 \times 1 = 0$

Max for TO = 0.10368 (From VB)

For NN,

From PRP: $0 \times 0.1 \times 0 = 0$

From VB: $0.3456 \times 0.4 \times 0 = 0$

From TO: $0 \times 0.5 \times 0 = 0$

From NN: $0 \times 0.1 \times 0 = 0$

Max for NN = 0

Finally, we calculate the probabilities for "play"

For PRP,

From PRP: $0 \times 0 \times 0 = 0$

From VB: $0 \times 0.3 \times 0 = 0$

From TO: $0.10368 \times 0.1 \times 0 = 0$

From NN: $0 \times 0.1 \times 0 = 0$

Max for PRP = 0

For VB,

From PRP: $0 \times 0.9 \times 0.2 = 0$

From VB: $0 \times 0 \times 0.2 = 0$

From TO: $0.10368 \times 0.4 \times 0.2 = 0.008294$

From NN: $0 \times 0.4 \times 0.2 = 0$

Max for VB = 0.008294 (From TO)

For TO,

From PRP: $0 \times 0 \times 0 = 0$

From VB: $0 \times 0.3 \times 0 = 0$

From TO: $0.10368 \times 0 \times 0 = 0$

From NN: $0 \times 0.1 \times 0 = 0$

Max for TO = 0

For NN,

From PRP: $0 \times 0.1 \times 0.8 = 0$

From VB: $0 \times 0.4 \times 0.8 = 0$

From TO: $0.10368 \times 0.5 \times 0.8 = 0.041472$

From NN: $0 \times 0.1 \times 0.8 = 0$

Max for NN = 0.041472 (From TO)

Therefore, we see that the highest probability for the word "play" is for the tag NN (0.041472).

Hence, the most likely POS tag for the word "play" in the sentence "He likes to play" is Noun (NN)

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Emission probabilities

	He	likes	to	play
PRP	0.8	0.2	0	0
VB	0	0.8	0	0.2
TO	0	0	1	0
NN	0.2	0	0	0.8

Transition probabilities

	VB	TO	NN	PRP
<s>	0.2	0	0.2	0.6
VB	0.0	0.3	0.4	0.3
TO	0.4	0	0.5	0.1
NN	0.4	0.1	0.1	0
PRP	0.9	0.0	0.1	0

Solution:

Start by calculating the probabilities for each POS tag (PRP, VB, TO, NN) based on the start state (<s>)

$V(\text{PRP}, ""\text{He} "") = P(\text{PRP}) \times P(""\text{He} "" \mid \text{PRP}) = 0.6 \times 0.8 = 0.48$

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$V(\text{TO}, ""\text{He} "") = P(\text{TO}) \times P(""\text{He} "" \mid \text{TO}) = 0 \times 0 = 0$

$V(\text{NN}, ""\text{He} "") = P(\text{NN}) \times P(""\text{He} "" \mid \text{NN}) = 0.2 \times 0.2 = 0.04$

Now, calculate the probabilities for "likes" using the results from "He"

For PRP,

From PRP: $0.48 \times 0 \times 0.2 = 0$

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From PRP: $0.48 \times 0.9 \times 0.8 = 0.3456$

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Max for VB = 0.3456 (From PRP)

For TO,

From PRP: $0.48 \times 0 \times 0 = 0$

From VB: $0 \times 0.3 \times 0 = 0$

From TO: $0 \times 0 \times 0 = 0$

From NN: $0.04 \times 0.1 \times 0 = 0$

Max for TO = 0

For NN,

From PRP: $0.48 \times 0.1 \times 0 = 0$

From VB: $0 \times 0.4 \times 0 = 0$

From TO: $0 \times 0.5 \times 0 = 0$

From NN: $0.04 \times 0.1 \times 0 = 0$

Max for NN = 0

Then, we calculate the probabilities for "to"

For PRP,

From PRP: $0 \times 0 \times 0 = 0$

From VB: $0.3456 \times 0.3 \times 0 = 0$

From TO: $0 \times 0.1 \times 0 = 0$

From NN: $0 \times 0 \times 0 = 0$

Max for PRP = 0

For VB,

From PRP: $0 \times 0.9 \times 0 = 0$

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From PRP: $0 \times 0 \times 1 = 0$

From VB: $0.3456 \times 0.3 \times 0.1 = 0.10368$

From TO: $0 \times 0 \times 1 = 0$

From NN: $0 \times 0.5 \times 1 = 0$

Max for TO = 0.10368 (From VB)

For NN,

From PRP: $0 \times 0.1 \times 0 = 0$

From VB: $0.3456 \times 0.4 \times 0 = 0$

From TO: $0 \times 0.5 \times 0 = 0$

From NN: $0 \times 0.1 \times 0 = 0$

Document 5 (d5): "Harry Ron Hermione friends"

b) Compute the importance score of every sentence using the sum of weights (normalized by size of pre-processed sentence) of all the words in the above preprocessed sentences. Use only the new simplified formula to compute word's weightage.

Weight (word W_i) = $TF(i,j) * N/DF(i)$

where,

N = Number of document given in the collection

$TF(i,j)$ = Frequency of word i in document j

$DF(i)$ = Number of documents containing word i

Solution:

Number of document given in the collection $N = 5$

$TF(i,j)$ = Frequency of word i in document j

$DF(i)$ = Number of documents containing word i

Word	TF(d1)	TF(d2)	TF(d3)	TF(d4)	TF(d5)	DF
Harry	1	1	0	0	1	3
brave	1	0	0	0	0	1
boy	1	0	0	1	0	2
shouted	0	1	1	0	0	2
Hermione	0	1	1	0	1	3
Expecto	0	0	1	0	0	1
Patronum	0	0	1	0	0	1
Ron	0	0	0	1	1	2
funny	0	0	0	1	0	1
friends	0	0	0	0	1	1

Document 1 (d1): Harry brave boy

Harry = $1 * 5/3 = 1.66$

brave = $1 * 5/1 = 5$

boy = $1 * 5/2 = 2.5$

Total Weight = $1.66+5+2.5 = 9.16$

Importance Score (normalized by 3 words): $9.16/ 3= 3.05$

Document 2 (d2): Harry shouted Hermione

Harry = $1 * 5/3 = 1.66$

shouted = $1 * 5/2 = 2.5$

Hermione = $1 * 5/3 = 1.66$

Total Weight = $1.66+5+1.66 = 5.82$

Importance Score (normalized by 3 words): $5.82 / 3 = 1.94$

Document 3 (d3): Hermione shouted Expecto Patronum

Hermione = $1 * 5/3 = 1.66$

shouted = $1 * 5/2 = 2.5$

Expecto = $1 * 5/1 = 5$

Patronum = $1 * 5/1 = 5$

Total Weight = $1.66 + 2.5 + 5 + 5 = 14.16$

Importance Score (normalized by 3 words): $14.16 / 4 = 3.54$

Document 4 (d4): Ron funny boy

Ron = $1 * 5/2 = 2.5$

funny = $1 * 5/1 = 5$

boy = $1 * 5/2 = 2.5$

Total Weight = $2.5 + 5 + 2.5 = 10$

Importance Score (normalized by 3 words): $10 / 3 = 3.33$

Document 5 (d5): Harry Ron Hermione friends

Harry = $1 * 5/3 = 1.66$

Ron = $1 * 5/2 = 2.5$

Hermione = $1 * 5/3 = 1.66$

friends = $1 * 5/1 = 5$

Total Weight = $1.66 + 2.5 + 1.66 + 5 = 10.82$

Importance Score (normalized by 4 words): $10.82 / 4 = 2.70$

Document Importance Score

d1 3.05

d2 1.94

d3 3.54

d4 3.33

d5 2.70

c) Extract summary in reverse chronological order with the top 2 most informative documents from part b)'s result.

Based on the importance scores calculated in part (b), the two most informative documents are:

Document 3 (d3) with a score of 3.54

Document 4 (d4) with a score of 3.33

Summary in Reverse Chronological Order:

Document 4 (d4): "Ron funny boy"

Document 3 (d3): "Hermione shouted Expecto Patronum"

d) Use ROUGE-1 to evaluate the results of the above system summary obtained in part b) w.r.t the below reference summary.

"Harry, Ron and Hermione are good friends"

Unigrams in System Summary:

Ron, funny, boy, Hermione, shouted, Expecto, Patronum

Unigrams in Reference Summary:

Harry, Ron, and, Hermione, are, good, friends

ROUGE-1 Precision:

Overlapping Unigrams between system summary and reference summary:

{Ron, Hermione}

ROUGE-1 Precision: $= 2/9 = 0.222$

ROUGE-1 Recall: $2/7 = 0.285$

ROUGE-1 F1-Score: $= 2 * 0.222 * 0.285 / (0.222 + 0.285)$

$= 0.126 / 0.507$

ROUGE-1 F1-Score: $= 0.248$

ROUGE-1 Precision:

$$\text{Precision} = \frac{\text{Number of overlapping unigrams}}{\text{Total unigrams in system summary}} =$$

ROUGE-1 Recall:

$$\text{Recall} = \frac{\text{Number of overlapping unigrams}}{\text{Total unigrams in reference summary}} =$$

ROUGE-1 F1-Score:

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Use Viterbi algorithm for a HMM tagger to tag the following sentence: The girl laughed

The transition and emission probabilities are as follows:

	N	V	DET
<s>	0.35	0	0.65
N	0.25	0.65	0.10
V	0.60	0.35	0.05
DET	0.65	0.35	0

	The	girl	laughed
N	0.0	1	0.0
V	0.0	0.0	1
DET	1	0	0

SOLUTION:

We start by initializing the probabilities for the first word "The" using the transition probabilities from the start state <s> and the emission probabilities. "The" of

Probabilities for : "The"

- $(\text{DET})P = \langle s \rangle \rightarrow (\text{DET} \times \text{DET})P \rightarrow (\text{The}) = 65.0 \times 1 = 65.0$
- $(\text{N})P = \langle s \rangle \rightarrow (\text{N} \times \text{N})P \rightarrow (\text{The}) = 35.0 \times 0 = 0$
- $(\text{V})P = \langle s \rangle \rightarrow (\text{V} \times \text{V})P \rightarrow (\text{The}) = 0 \times 0 = 0$

We now calculate the probabilities for the second word "girl" based on the previous state probabilities and transitions

Probabilities for : "girl"

For N : (Noun)

$(\text{N})P_{\max} = ((\text{DET})P \times \text{DET})P \rightarrow (\text{N} \times \text{N})P \rightarrow (\text{"girl"})P = (\text{N} \times \text{N})P \rightarrow (\text{"girl"})P \times (\text{V})P \rightarrow (\text{N} \times \text{N})P \rightarrow (\text{"girl"})$

$P(\text{N}) = \max(0.65 \times 65.0, 10 \times 0.25 \times 1, 0 \times 60.0 \times 1) = 42.25$

For V (Verb):

$$P(V)=\max((DET)P \times (DET)P \rightarrow (V \times V)P \rightarrow, ("girl" P(N) \times P(N \rightarrow V) \times P(V \rightarrow "girl"), (V)P \times V)P \rightarrow (V \times V)P \rightarrow ("girl" (V)P=65.0) \max \times 35.0 \times, 00 \times 0.65 \times 0, 0 \times 35.0 \times (0=0$$

ForDET :(Determiner)

$$P(DET)=\max((DET)P \times (DET)P \rightarrow (DET \times DET)P \rightarrow, ("girl" P(N) \times P(N \rightarrow DET) \times P(DET \rightarrow "girl"), (V)P \times V)P \rightarrow (DET \times DET)P \rightarrow g("irl) \\ = (DET)P \max (0.65 \times 0 \times, 00 \times 0.10 \times 0, 0 \times 05.0 \times (0=0$$

Wenow compute the probabilities for the third word ."laughed"

Probabilitiesf or:"laughed"

ForN :(Noun)

$$=(N)P \max ((N)P \times N)P \rightarrow (N \times N)P \rightarrow, ("laughed" P(V) \times P(V \rightarrow N) \times P(N \rightarrow "laughed"), (DET)P \times (DET)P \rightarrow (N \times N)P \rightarrow laughe("d) \\ 2275.0) \max = (N)P \times 25.0 \times, 00 \times 60.0 \times, 00 \times 0.65 \times 0) = 0$$

ForV :(Verb)

$$P(V) \max = ((N)P \times N)P \rightarrow (V \times P(V \rightarrow, ("laughed" P(V) \times P(V \rightarrow V) \times P(V \rightarrow "laughed"), (DET)P \times (DET)P \rightarrow (V \times V)P \rightarrow "laughed" () \\ P(V)=\max(0.4225 \times 65.0 \times, 10 \times 0.35 \times 1, 0 \times 35.0 \times (1=274625.0$$

ForDET :(Determiner)

$$P(DET)=\max((N)P \times N)P \rightarrow (DET \times DET)P \rightarrow, ("laughed" P(V) \times P(V \rightarrow DET) \times P(DET \rightarrow "laughed"), (DET)P \times (DET)P \rightarrow (DET \times DET)P \rightarrow ("laughed") \\ P(DET)2275.0) \max = \times 0.10 \times 0, 0 \times 05.0 \times, 00 \times 0 \times (0=0$$

FinalTagged Sequence:

The/DET N/girl V/laughed

Question 3. [6 Marks]

Consider the statement: "The extremely hot desert is a hostile place for living"

Using Depth First, Simple Top Down Parsing method, work out the steps only till the phrase “The extremely hot desert” is successfully parsed. Provide one line explanation for each step.

The grammar rules to be applied and the lexicon are given below.

Grammar	Word	Category	Word	Category
$S \rightarrow NP VP$	The	DET	a	DET
$NP \rightarrow DET AP N$	extremely	ADV	hostile	ADJ
$NP \rightarrow DET N$	hot	ADJ	place	N
$AP \rightarrow ADV ADJ$	desert	N	for	PREP
$AP \rightarrow ADJ$	is	V	living	N ADJ
$VP \rightarrow V NP$				
$VP \rightarrow V NP PP$				
$PP \rightarrow PREP NP$				

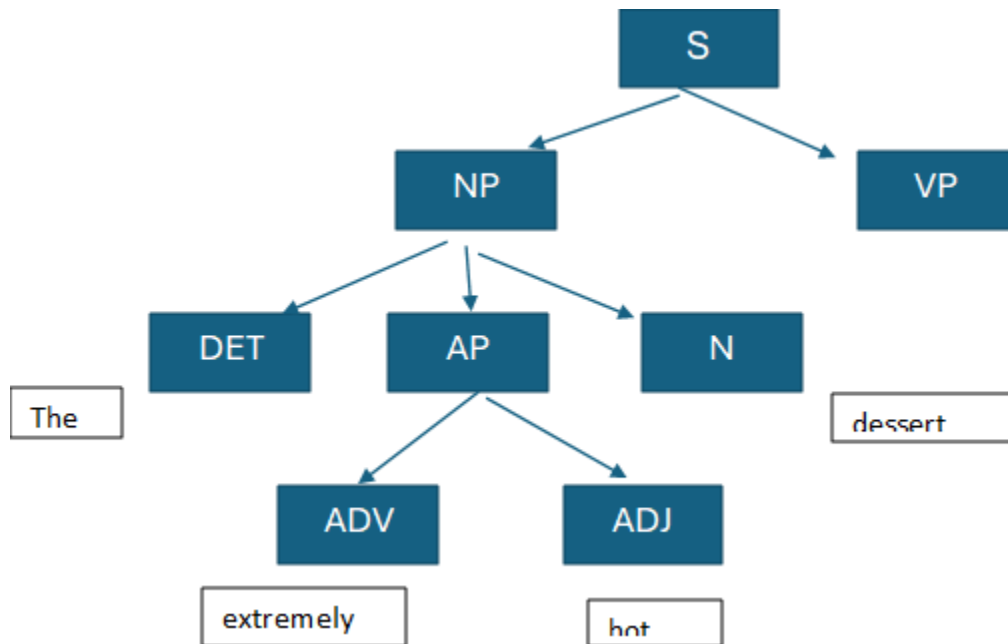
Solution:

1The 2extremely 3hot 4desert 5is 6a 7hostile 8place 9for 10living11

Step	Actual State	Backup State	Comments
1	((SP)1)		
2	((NP VP)1)		
3	((DET AP N VP) 1	((DET N)1)	
4	((AP N VP)2)	((DET N)1)	'The' is Det
5	((ADV ADJ N VP)2)	((ADJ N VP) 2) ((DET N)1)	AP -> ADV ADJ AP -> ADJ is put to backup state
6	((ADJ N VP)3)	((ADJ N VP) 2) ((DET N)1)	'extremely' is ADV
7	((N VP)4)	((ADJ N VP) 2) ((DET N)1)	'hot' is ADJ

8	((VP)5)	((ADJ N VP) 2) ((DET N)1)	'desert' is N

Parse Tree:



Question 5. [3+3=6 Marks]

- a) Draw the knowledge graph that shows an example of a derived inference. 3 marks

RDF Statements:

ex:LivingThing rdfs:subClassOf ex:Object

ex:Animal rdfs:subClassOf ex:LivingThing

ex:Person rdfs:subClassOf ex:Animal

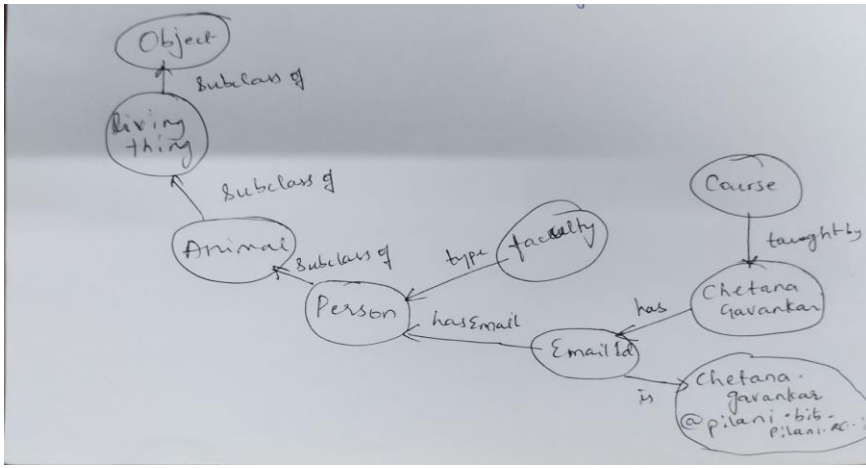
ex:Faculty rdf:type ex:Person

ex:Person rdf:hasEmail ex:EmailId

ex:Course rdf:taughtBy "Chetana Gavankar"

ex:"Chetana Gavankar" rdf:EmailId ex:"chetana.gavankar@pilani.bits-pilani.ac.in"

Solution: (Sample Knowledge graph)



- b) Consider the following sentence: "Time flies like an arrow through the air." The following entries are available in a dictionary for the word "flies": 3 marks

Word	Gloss	Usage Examples
flies (1) V	move through the air using wings	Hummingbirds flies like a helicopter.
flies (2) N	a winged insect	Banana flies sit on fruits.

We need to disambiguate the word "flies" in the above sentence, using the Lesk algorithm. Explain the steps to be followed.

Solution :

Identifying the overlaps - 1M

Explain the Steps - 1M

Identifying the correct Gloss - 1M

Gloss 1 (Verb): "move through the air using wings"

The gloss contains the word air, which also appears in the sentence ("through the air").

Gloss 2 (Noun): "a winged insect"

There are no words in the gloss (like insect or fruits) that match the context of the sentence.

Select the sense with the highest overlap:

Gloss 1 (Verb) has a word (air) that matches the context of the sentence, while Sense 2 (Noun) has no overlapping terms.

Hence, Gloss 1 is selected as the correct interpretation: "flies" as a verb, meaning to move through the air using wings.

Question 6. [2 + 4 = 6 Marks]

- a) Consider following contextual embeddings for the word "bank" for each of its senses:

bank _N ¹	Financial institution	[1,4,1]
bank _N ²	Sloping land	[2,8,1]

bank _v ¹	Tip laterally	[4,7,3]
bank _v ³	Do business with a bank	[3,3,3]

The contextual embedding for the target word “bank”

in a specific sentence was derived as [2,6,2]. What is the sense of the target word?

Solution:

We find the cosine similarity between target word and each sense.

Cosine Similarity = Dot Product of Vectors / Product of magnitude of each vector

Sense				Magnitude	Target				Target Magnitude				Dot Product	Product of Magnitudes	Cosine Similarity
	1	4	1	4.24	2	6	2		6.633249581				28	28.14	0.995
	2	8	1	8.31									54	55.10	0.980
	4	7	3	8.60									56	57.06	0.981
	3	3	3	5.20									30	34.47	0.870

Hence the Answer is sense “Financial Institution” or bank_N¹(2 marks)

b) Given the sentence: “Stay Curious” and information below, demonstrate the mathematical application of a single attention head to derive Self Attention. Support your answer with diagrams and relevant mathematical equations, detailing all steps. Apply score normalization, masking and softmax.

Embeddings:

	Embedding
Stay	[1 1]
Curious	[1 1]

Weight Matrix for Query W^q

[1 0

0 2]

Weight Matrix for Keys W^k

[0 2

1 0]

Weight Matrix for Values W^v

[2 3

3 2]

- $\text{Exp}(3.54) = 34.31$
- Compute all fractions only to Second Decimal place (X.XX)

Solution:

(Marking scheme in brackets against each step)

1. (0.5 marks)

	Input
Stay	[1 1]
Curious	[1 1]

2. Compute Q, K & V (1.5 marks)

Q

1 2

1 2

K

1 2

1 2

V

5 5

5 5

3. Compute QK^T (0.5 marks)

5	5
5	5

4. Apply Normalization with $d_k=2$ (0.5 marks)

3.54	3.54
3.54	3.54

5. Apply Masking (0.5 marks)

3.54	3.54
3.54	3.54

6. Apply Softmax (0.5 marks)

formula for softmax

1.00	0.00
0.50	0.50

7. Multiply above Weights with V (Final Answer) (0.5 marks)

Self-Attention =

5	5
5	5